DUNMAN SECONDARY SCHOOL

CANDIDATE NAME	

CLASS		INDEX NUMBER	

PRELIMINARY EXAMINATION 2023
SECONDARY 4 EXPRESS

MATHEMATICS 4052/02

Paper 2

24 August 2023

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Calculator Model

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.
If working is needed for any question it must be shown with the answer.
Omission of essential working will result in loss of marks.
The total of the marks for this paper is 90.

The use of an approved scientific calculator is expected, where appropriate.
If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For π, use either your calculator value or 3.142, unless the question requires the answer in terms of π.

Total Marks

Mathematical Formulae

Compound interest

$$\text{Total amount} = P\left(1+\frac{r}{100}\right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi rl$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2}ab\sin C$$

$$\text{Arc length} = r\theta\text{, where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2}r^2\theta\text{, where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$$

Answer **all** the questions.

1 **(a)** Solve these simultaneous equations.

$$7x + 8y = 14$$
$$x + 4y = 12$$

You must show your working.

Answer $x =$ ……………………….

Answer $y =$ ………………………. [3]

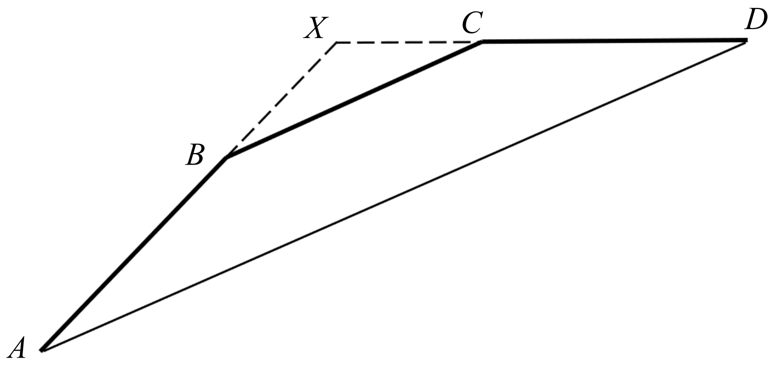
(b) Simplify $\dfrac{8x^2}{27y^3} \div \dfrac{4}{3x^3y}$.

Answer …………………………. [2]

(c) Simplify $\dfrac{2x^2 - 18}{x^2 - 6x + 9}$.

Answer …………………………. [2]

2



The diagram shows vertices A, B, C and D of a regular n-sided polygon.
The point X is extended from AB and DC.
AD is parallel to BC.

(a) Explain why triangle XBC is an isosceles triangle.

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………

………………………………………………………………………………………… [2]

(b) It is given that angle $ABC = 156°$.

Find

(i) the value of n,

Answer $n =$ ……………………… [2]

(ii) angle BXC,

Answer ……………………… ° [1]

(iii) angle CBD,

Answer ……………………… ° [1]

(iv) angle AYD, given that Y is another vertex of the polygon.

Answer ……………………… ° [3]

3 The first four terms in a sequence of numbers are given below.

$T_1 = 5 \times 1 = 5$
$T_2 = 7 \times 5 = 35$
$T_3 = 9 \times 9 = 81$
$T_4 = 11 \times 13 = 143$

(a) Find T_5 .

Answer .... [1]

(b) Show that the nth term of the sequence, T_n, is given by $8n^2 + 6n - 9$.

Answer

[3]

(c) Explain why the value of T_n cannot be even for all values of n.

....

.... [1]

(d) T_p and T_{p+1} are consecutive terms in the sequence.
Find and simplify an expression, in terms of *p*, for $T_{p+1} - T_p$.

Answer .... [3]

(e) Explain why the difference between any two consecutive terms in the sequence is always even.

....

.... [1]

4 **(a)** Complete the table of values for $y = -\frac{1}{4}x^3 + x + 1$.

x	–4	–3	–2	–1	0	1	2	3	4
y		4.75	1	0.25	1	1.75	1	–2.75	–11

[1]

(b) On the grid opposite, draw the graph of $y = -\frac{1}{4}x^3 + x + 1$ for $-4 \le x \le 4$. [3]

(c) By drawing a tangent, find the gradient of the curve at (3, –2.75).

Answer .... [2]

(d) The equation $x^3 + 4x - 8 = 0$ can be solved by finding the points of intersection of the straight line $y = ax + b$ and the curve $y = -\frac{1}{4}x^3 + x + 1$.

(i) Find the value of a and the value of b.

Answer $a =$ ....

$b =$ .... [2]

(ii) By drawing the line $y = ax + b$, solve the equation $x^3 + 4x - 8 = 0$.

Answer $x =$ .... [2]

(e) State the values of k for which $-\frac{1}{4}x^3 + x + 1 = k$ has exactly two solutions.

Answer $k =$ .... or .... [2]

5 A chef and his apprentice are preparing dumplings for a dinner party.

(a) The chef can wrap 15 dumplings in x minutes.

Write an expression, in terms of x, for the number of dumplings that the chef wraps in one minute.

Answer ………………………… [1]

(b) His apprentice takes 1 minute more to wrap 15 dumplings.

Write an expression, in terms of x, for the number of dumplings that his apprentice wraps in one minute.

Answer ………………………… [1]

(c) The chef and his apprentice took 1 hour to wrap 960 dumplings.

Write down an equation in x to represent this information and show that it reduces to

$$16x^2 - 14x - 15 = 0 \ .$$

Answer

[3]

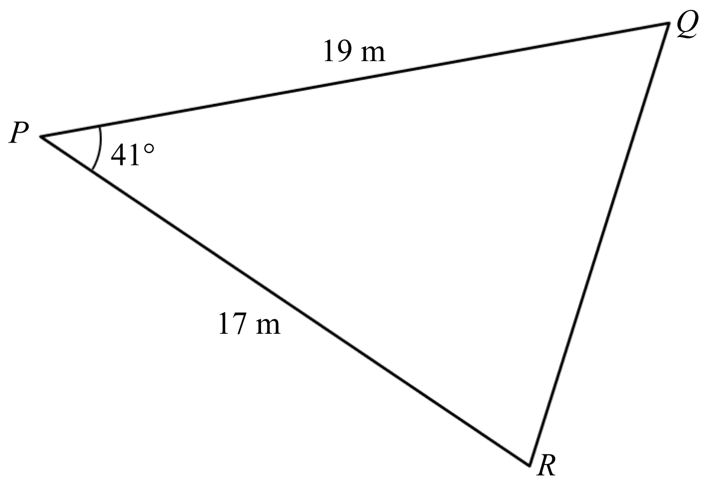
(d) Solve the equation $16x^2 - 14x - 15 = 0$.

Answer $x =$ ……………… or $x =$ ……………… [3]

(e) Find the number of dumplings that his apprentice can wrap in 1 hour.

Answer ………………………… [2]

6



The diagram shows a field PQR on horizontal ground.
$PQ = 19$ m, $PR = 17$ m and angle $QPR = 41°$.

(a) Calculate QR.

Answer ……………………… m [3]

(b) Calculate angle PRQ.

Answer ……………………… ° [2]

(c) A drone is hovering at 5.2 m vertically above *P*.

Find the angle of elevation from *Q* to the drone.

Answer ……………………… ° [2]

(d) The drone flies along the line *PQ* while maintaining a height of 5.2 m above the ground. The drone then stops at a point with the largest angle of depression from the drone to *R*.

Find the horizontal distance of the drone from *R* at this point.

Answer ……………………… m [2]

7 **(a)** The masses of a set of 20 bolts from Company *A* was recorded.
Their masses are shown in the stem-and-leaf diagram.

0	8	9	9							
1	2	2	2	4	4	6	8	9	9	9
2	0	1	1	3	3	3	4			

Key: 1 | 2 means 1.2 grams

(i) The acceptable mass for a bolt is between 1.1 g and 2.2 g.
Find the percentage of bolts that are not accepted.

Answer ……………………… % [1]

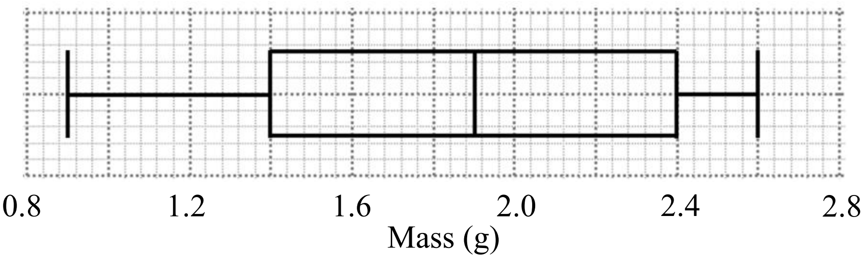
(ii) Find the median mass.

Answer ……………………… g [1]

(iii) Find the interquartile range.

Answer ……………………… g [2]

(iv) Jason wants to replace the bolts in his model aircraft with lighter bolts.
The box-and-whisker plot shows the distribution of masses of bolts from Company *B*.



Suggest which company Jason should buy his bolts from.
Make two justifications for the decision you make.

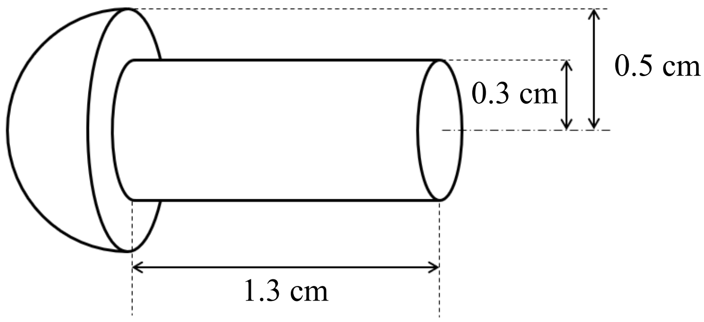
……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

…………………………………………………………………………………… [2]

(b)



The diagram shows a bolt, which is made of a hemisphere and cylinder.

Calculate

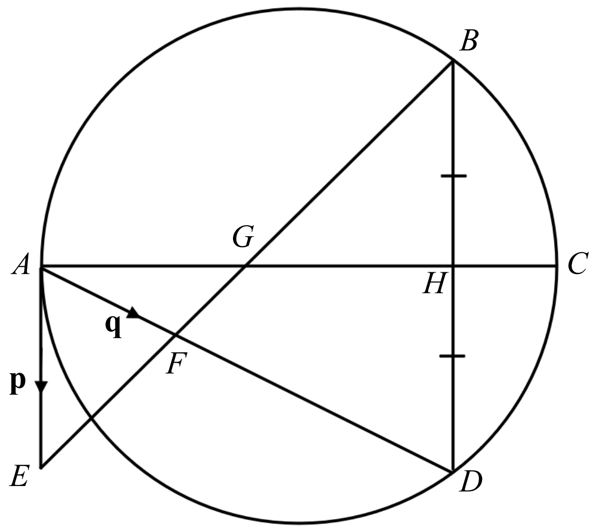
(i) the volume of the bolt,

Answer ……………………… cm^3 [3]

(ii) the surface area of the bolt.

Answer ……………………… cm^2 [3]

8



The diagram shows a circle with diameter AC.
AE is a tangent to the circle.
BE intersects AD and AC at F and G respectively.
AC bisects BD at H and $AF : FD = 1 : 2$.
$\overrightarrow{AE} = \mathbf{p}$ and $\overrightarrow{AF} = \mathbf{q}$.

(a) Explain why AE is parallel to BD.

....

....

....

.... [3]

(b) Show that triangles AFE and DFB are similar.
Give a reason for each statement you make.

....

....

.... [2]

(c) Express, in terms of **p** and **q**, as simply as possible,

(i) $\overrightarrow{AH}$,

Answer .... [2]

(ii) $\overrightarrow{AG}$.

Answer .... [2]

(d) Find the ratio of the area of triangle *AFE* to the area of triangle *AFG*.

Answer .... [3]

9 An Air Medical Service (AMS) supports 2 towns, X and Y, which are 135 km apart.
A scale diagram of X and Y is shown on the next page.

Brian is an AMS air navigation officer who studies the flight data of the helicopter used by the AMS.

He obtains the following average speed data of the helicopter in the last 100 flights.

Average speed, (s km/h)	$140 \le s < 160$	$160 \le s < 180$	$180 \le s < 200$	$200 \le s < 220$	$220 \le s < 240$
Frequency	3	16	25	38	18

(a) Calculate an estimate of the mean and standard deviation.

Answer Mean ………………………………. km/h

Standard deviation ………………… km/h [2]

(b) Brian did some statistical research and found that 95% of the average speed of all helicopter flights will be between p km/h to q km/h, where

p = [mean in **(a)**] – 2×[standard deviation in **(a)**] ,
q = [mean in **(a)**] + 2×[standard deviation in **(a)**] .

Show that 95% of all helicopter flights from town X to town Y will take at most 0.853 h.

Answer

[2]

(c) Brian is tasked to look at relocating the AMS in order to support an additional town Z.
Z is on a bearing of 130° from X and on a bearing of 210° from Y.
The new location of the AMS needs to allow the AMS to support each town in the shortest amount of time.
Brian thinks that at the new location, the AMS helicopter will take at most 15 minutes to reach any town.

By marking on the diagram the new position of the AMS with the point P, determine whether Brian is correct.
Justify your decision with calculations.

[6]

Continuation of working space for question 9.

END OF PAPER